



Lecture 13 Handling network data 10 Dec 2025

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Announcements

- Final project milestone P3 due on **Sun 21 Dec** 23:59
- Friday's lab session:
 - Project office hour (on Zoom)
 - Last project office hour before due date
 - To secure your team's personal time slot, follow protocol described in [this Ed post](#)
 - Exercises on handling networks in CE 14 ([link](#))
 - In parallel to office hour

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Feedback

Give us feedback on this lecture here:
<https://go.epfl.ch/ada2025-lec13-feedback>

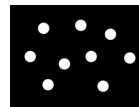
- What did you (not) like about this lecture?
- What was (not) well explained?
- On what would you like more (fewer) details?
- ...

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Beyond flat tables

People

id	name	age
1	Bob	36
2	Willy	32
...	...	...



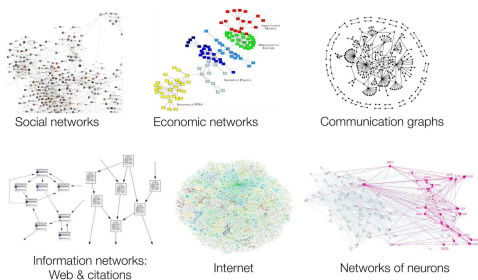
Marriages

husband_id	wife_id
1	34
2	5
2	87
...	...



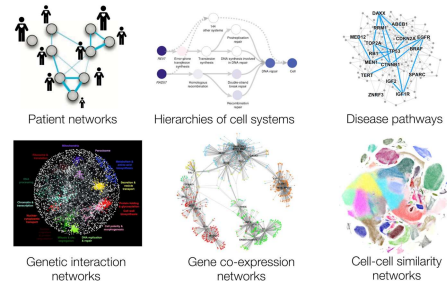
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Examples



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Examples



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Networks as graphs

- **Network:** a real-world system of dependent variables, e.g.,
 - WWW is a network of hyperlinked documents
 - Society is a network of individuals linked by family, friendship, professional ties
 - Metabolic network is sum of all chemical reactions in a cell
- **Graph:** mathematical abstraction for describing networks
- In practice, “network” and “graph” are used interchangeably
- You can make a graph out of almost anything (e.g., connect all people whose name starts with the same letter), so must ask: Does this graph correspond to a meaningful network?

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Today's lecture

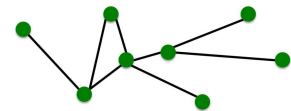
- **Part 1:** Types of graphs
- **Part 2:** Representing graphs on computers
- **Part 3:** Properties of real-world networks
- **Part 4:** Measuring node importance

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Part 1: Types of graphs

Most basic type: undirected graphs

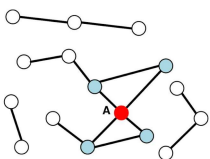
- Entities:
nodes/vertices V
- Relationships/interactions:
edges/links E
- Entire system:
graph $G = (V, E)$



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Node degree



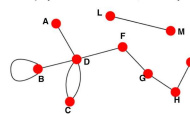
Node degree, k_i : the number of edges adjacent to node i
 $k_A = 4$

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Types of graphs: undirected vs. directed

Undirected

- **Links:** undirected (symmetrical, reciprocal)

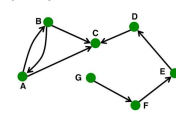


Examples:

- Collaborations
- Friendship on Facebook

Directed

- **Links:** directed (arcs)

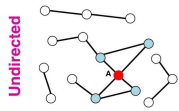


Examples:

- Phone calls
- Following on Twitter

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Average node degree



Node degree, k_i : the number of edges adjacent to node i
 $k_A = 4$

Avg. degree: $\bar{k} = \langle k \rangle = \frac{1}{N} \sum_{i=1}^N k_i =$



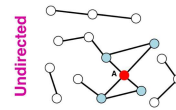
POLLING TIME

Scan QR code or go to
<https://go.epfl.ch/ada2025-lec13-q>
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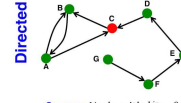
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Average node degree



Node degree, k_i : the number of edges adjacent to node i
 $k_A = 4$

Avg. degree: $\bar{k} = \langle k \rangle = \frac{1}{N} \sum_{i=1}^N k_i = \frac{2E}{N}$



In directed networks we define an **in-degree** and **out-degree**.

The (total) degree of a node is the sum of in- and out-degrees.

$k_C^{in} = 2$ $k_C^{out} = 1$ $k_C = 3$

$\bar{k} =$

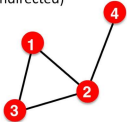
Source: Node with $k^{in} = 0$

Sink: Node with $k^{out} = 0$

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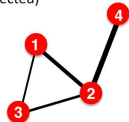
Types of graphs: weighted

- **Unweighted** (undirected)



Examples: Friendship, Hyperlink

- **Weighted** (undirected)

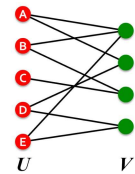


Examples: Collaboration, Internet, Roads

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Types of graphs: bipartite

- **Bipartite graph** is a graph whose nodes can be divided into two disjoint sets U and V such that every link connects a node in U to one in V



- **Examples:**

- Authors-to-Papers (they authored)
- Actors-to-Movies (they appeared in)
- Users-to-Movies (they rated)
- Recipes-to-Ingredients (they contain)

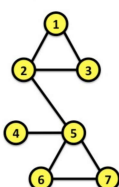
- **"Folded" networks (a.k.a. "projections"):**

- Author collaboration networks
- Movie co-rating networks

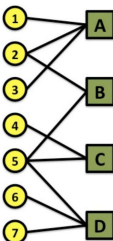
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Projections of bipartite graph

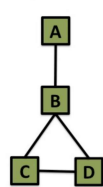
Projection U



U V

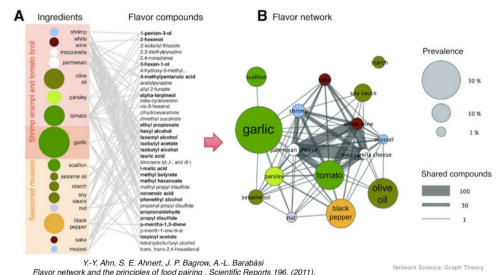


Projection V



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Example: flavor networks



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Part 2: Representing graphs on computers

Representing graphs on computers: adjacency matrix



$A_{ij} = 1$ if there is a link from node i to node j
 $A_{ij} = 0$ otherwise

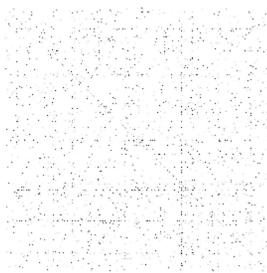
$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \quad A = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Note that for a directed graph (right) the matrix is not symmetric.

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Adjacency matrix usually sparse



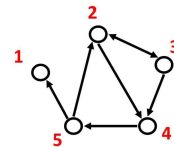
What is a better representation? 🤔

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Representing graphs on computers: edge list

Represent graph as a set of edges:

- (2, 3)
- (2, 4)
- (3, 2)
- (3, 4)
- (4, 5)
- (5, 2)
- (5, 1)

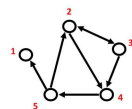


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Representing graphs on computers: adjacency list

Adjacency list:

- Easier to work with if network is
 - Large
 - Sparse
- Allows us to quickly retrieve all neighbors of a given node
 - 1:
 - 2: 3, 4
 - 3: 2, 4
 - 4: 5
 - 5: 1, 2



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Graph processing libraries

- Good overview + benchmark available [here](#)
- NetworkX
 - Written in Python
 - Popular but slow
 - Ok when your graph is small
 - **This Friday's lab session: intro to NetworkX**
- NetworkKit, SNAP, iGraph, graph-tool
 - Written in C++
 - Much faster than NetworkX
 - Libraries also available in other languages (incl. Python)
 - Consider these if your graph is large

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Part 3:

- Real networks are different from arbitrary graphs
- Real networks tend to share certain properties
- Remarkable, given the diversity of networks
 - Information networks (e.g., Web graph, knowledge graphs)
 - Social networks (e.g., Facebook, dating networks)
 - Biological networks (e.g., protein–protein interaction)
 - ...

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sparsity

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Today!
And happy ADAvent
to all of you!



Random wiki fact:

Gemiler Island

Article Talk

From Wikipedia, the free encyclopedia

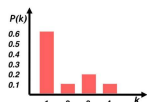
Gemiler Island (**Turkish**: *Gemiler Adası* or *Gemiler* → *Greek*: *Τεμάλις*) is an island located off the coast of *Turkey* near the city of *Adana*. On the island are the remains of several churches built between the fourth and sixth centuries AD, *signs* with a variety of associated buildings. Archaeologists believe the location of the original *Temple of Saint Nicholas* (the original Turkish name is *Gemile* from the Greek word *καμύλα* (*kamyla*) meaning *camel*, so named because of its geographical shape (see *Fevo Goniou Island Archaeological Site*)).

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degree distribution

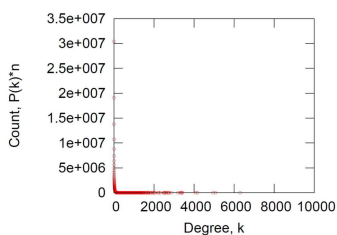
- $$N_k = \# \text{ nodes with degree } k$$

-



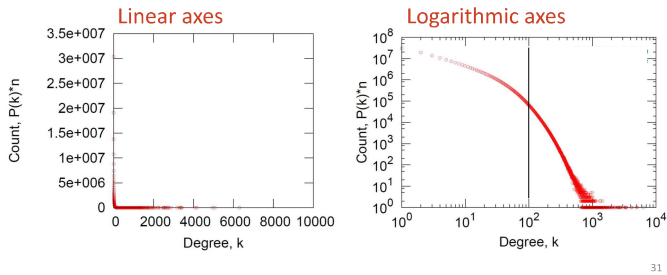
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degree distribution



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Properties of real-world networks: degree distribution

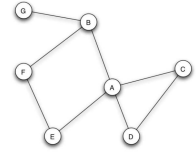


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Properties of real-world networks: triadic closure

- “A friend of my friend is my friend”
- Measured via clustering coefficient C_i of node i :

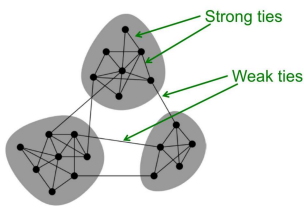
$$C_i = \frac{\text{\#edges among neighbors of } i}{\text{\#potential edges among neighbors of } i}$$
- #potential edges among neighbors of i in undirected graph: $k_i(k_i - 1) / 2$
- $C_A = 1 / (4 * 3 / 2) = 1/6$
- In real networks, nodes have **large clustering coefficients**



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Properties of real-world networks: community structure

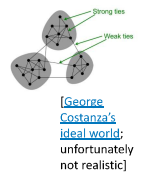
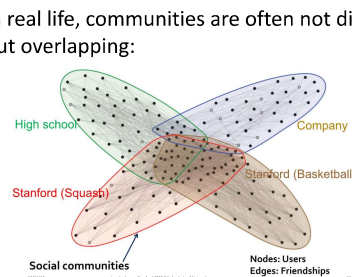
- Triadic closure makes real networks **cluster into locally dense “communities”**
- Communities connected via “weak ties”
- “[The strength of weak ties](#)” (Granovetter 1973)
- Weak ties fill “structural holes”



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Properties of real-world networks: community structure

- In real life, communities are often not disjoint, but overlapping:



[George Costanza's ideal world: unfortunately not realistic]

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Extra homework:
Start watching
Seinfeld,

so you understand the
[“Worlds Collide”](#)
[theory](#)

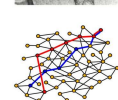
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Properties of real-world networks: average shortest-path length

- What is the typical shortest path length between any two people?
 - Experiment on the global friendship network
 - Can't measure, need to probe explicitly
- Small-world experiment [Milgram '67]
 - Picked 300 people in Omaha, Nebraska and Wichita, Kansas
 - Ask them to get a letter to a stock-broker in Boston by passing it through friends
- How many steps did it take?



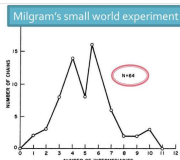
[Movie]



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Properties of real-world networks: average shortest-path length

- **64 chains completed:**
(i.e., 64 letters reached the target)
 - It took 6.2 steps on the average, thus
“6 degrees of separation”
- **Further observations:**

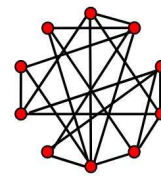


- ### Further observations:
- People who owned stock had shorter paths to the stockbroker than random people: 5.4 vs. 6.7
 - People from the Boston area have even closer paths: 4.4

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Properties of real-world networks: navigability

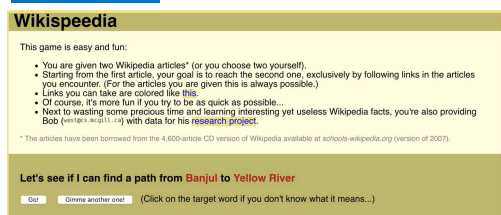
- For decades, people focused on the fact that short paths exist in social networks
 - But this is true even in random graphs (e.g., [Erdős-Rényi model](#))
 - The truly amazing fact is not that short paths exist, but that they are discoverable via greedy decentralized routing (as in Milgram's experiment)
 - Intrigued? Read Jon Kleinberg: ["The Small-World Problem: An Algorithmic Perspective"](#)
- 



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Don't believe me?

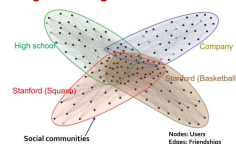
- Play a game and see for yourself how well you can navigate an a-priori unknown network:
 - [Wikispedia.net](https://www.wikispedia.net/)



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Properties of real-world networks: homophily/heterophily

- “Birds of a feather flock together”
 - Especially in social networks
 - Big confound and cause of debate: Influence vs. homophily
 - E.g., obese people’s friends are more likely to also be obese
 - Influence: I copy eating behavior of those around me [\[argument for influence\]](#)
 - Homophily: people with similar eating behavior prone to become friends [\[argument for homophily\]](#)
- 
- A Venn diagram illustrating the relationship between three groups: High school (green), Stanford (red), and Social communities (blue). The intersection of all three groups is labeled 'Nodes: User' and 'Edges: Friend'. The diagram shows that individuals who are in all three groups are the most likely to be friends, as indicated by the dense network of lines connecting them.



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Properties of real-world networks: summary

Real-world networks (across many types)

- are sparsely connected,
- but some nodes are much more connected than most others (i.e., skewed degree distribution);
- form locally dense clusters via triadic closure,
- which leads to community structure;
- have short paths between random node pairs (partly due to “hubs” [skewed degree distribution!]),
- and the short paths are easily discoverable.

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Part 4: Measuring node importance

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How to measure “importance” of a node?

- Formalized via [centrality measures](#)
- Map each node i to a scalar value $C(i)$ capturing its importance in the overall network

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Degree centrality

- Simplest centrality measure
- Many neighbors → important node
 - $C(i)$ = number of neighbors of i
- Very brittle, easy to “rig”
 - E.g., Twitter: scam account, followed by 100,000 other scam accounts

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Closeness centrality

- Farness(x) = total distance to x from other nodes
- $C(x) = 1 / \text{Farness}(x)$
 - Reciprocal to turn farness into closeness
 - Only defined for connected graphs (otherwise d infinite)
- Under **closeness centrality**, nodes that are easy to reach from anywhere in the network are considered important
- Variant: **harmonic centrality**: switch sum and reciprocal
 - $C(x)$ = total reciprocal distance of x to other nodes
 - Defined even for disconnected graphs (define $1/d$ of disconnected nodes as 0)

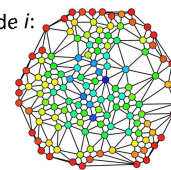
$$C(x) = \frac{1}{\sum_{y \neq x} d(y, x)}$$

$$C(x) = \sum_{y \neq x} \frac{1}{d(y, x)}$$

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Betweenness centrality

- $C(i)$ = average fraction of all shortest paths in the network that pass through node i
- Computation of betweenness centrality of node i :
 - For each pair of vertices (s, t) :
 - Find all shortest paths from s to t
 - Compute the fraction of these shortest paths that pass through i
 - Average this fraction over all pairs of vertices (s, t)
- Expensive to compute



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Katz centrality

- Generalization of degree centrality
- Degree centrality counts only number of direct neighbors (i.e., neighbors at distance 1)
- Katz centrality also counts neighbors at distances 2, 3, ...
- More precisely, number of paths from other nodes to i , giving less weight to larger distances:

$$C(i) = \sum_{k=1}^{\infty} \sum_{j=1}^N \alpha^k (A^k)_{ji}$$

k -th power of adjacency matrix A contains number of length- k paths for each node pair

- More robust than degree centrality

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PageRank centrality

- Recursive definition: my centrality $C(i) =: x_i$ is high if I receive inlinks from many other central nodes:



$$x_i = \sum_j a_{ji} \frac{x_j}{L(j)}$$

$$L(j) = \sum_i a_{ji}$$

a_{ji} : entry (j, i) of adjacency matrix A (1 if j links to i , else 0)
 $L(j)$: out-degree of j

- Some extra tweaks to make it work with any graph (e.g., disconnected)

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PageRank centrality

$$x_i = \sum_j a_{ji} \frac{x_j}{L(j)}$$

- Matrix notation: $\mathbf{x} = \mathbf{M} \mathbf{x}$
(where $\mathbf{M}$ is computed from adjacency matrix A)
- Do you recognize this?
 - $\mathbf{x}$ is eigenvector of $\mathbf{M}$ with eigenvalue 1 $\rightarrow$ we're in linear-algebra land (home sweet home)
 - $\mathbf{x}$ is the steady state the Markov chain induced by the network: x_i is fraction of time a random walker will have spent in node i , after a very long ($\rightarrow \infty$) random walk

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PageRank centrality



- The technology that made Google huge
 - "Page" in PageRank for Larry Page
 - MapReduce was invented to compute PageRank on full Google Web crawl
 - "The \$25,000,000,000 eigenvector" [\[link\]](#)
- Bottom line:
Pay attention in your linear algebra class

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Feedback

Give us feedback on this lecture here:
<https://go.epfl.ch/ada2025-lec13-feedback>

- What did you (not) like about this lecture?
- What was (not) well explained?
- On what would you like more (fewer) details?
- ...

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Credits

- Some slides borrowed from [CS224W class](#) (Jure Leskovec, Stanford)

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